

IoT Internship Report

Task 2 Sensor Data Simulation

Submitted By:

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Programme : Bachelor of Computer Applications (BCA) - IoT

INTRODUCTION

What is Sensor Data Simulation :

Sensor Data Simulation is the process of generating and monitoring sensor readings in a virtual environment. It helps developers understand how sensors collect, process, and transmit data without requiring physical hardware. In this task, a DHT22 sensor was simulated using the Wokwi platform to monitor temperature and humidity values.



COMPONENTS USED :

1. ESP32 Development Board

The ESP32 is a microcontroller used in IoT projects. It has built-in Wi-Fi and Bluetooth capabilities, allowing it to communicate with sensors and other devices. In this project, it receives and processes data from the DHT22 sensor.

2. DHT22 Temperature and Humidity Sensor

The DHT22 is a digital sensor that measures temperature and humidity. It collects environmental data and sends the readings to the ESP32 for monitoring and display.

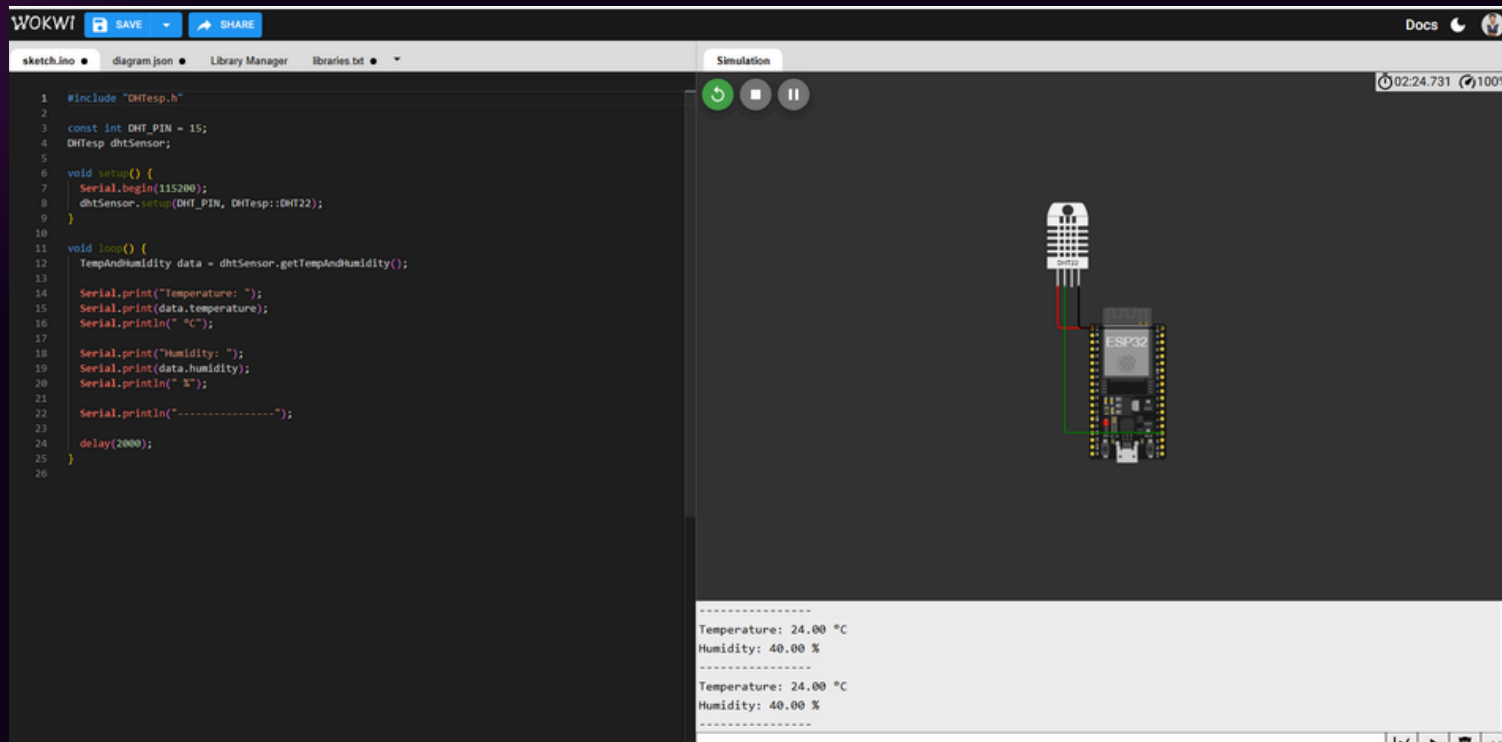
3. Wokwi IoT Simulator

Wokwi is an online IoT simulator that allows users to test and run electronic circuits virtually. It was used in this project to simulate the ESP32 and DHT22 sensor and monitor temperature and humidity data.

4. Arduino Programming Language

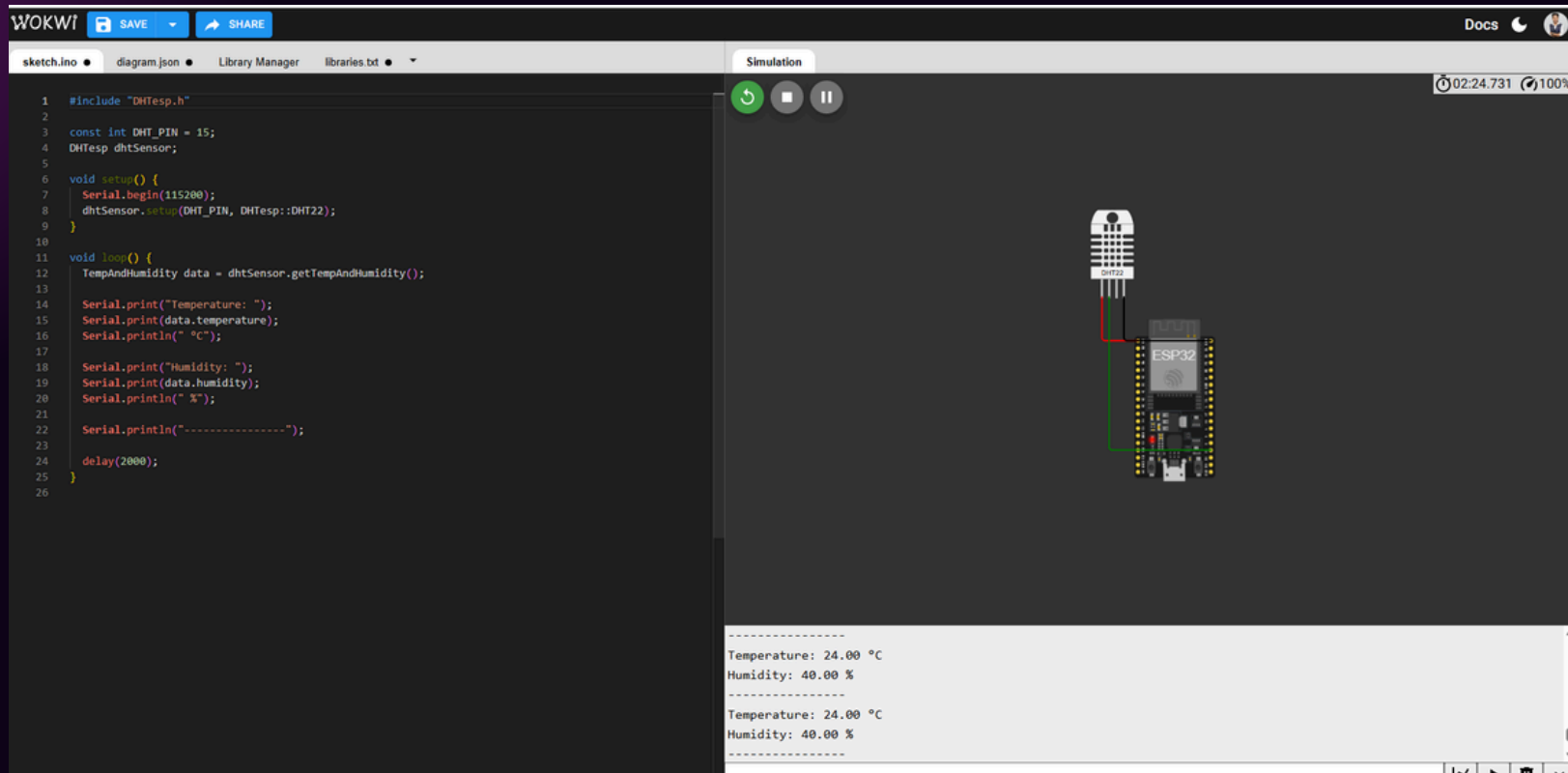
Arduino Programming Language was used to program the ESP32 and read temperature and humidity data from the DHT22 sensor.

CIRCUIT DIAGRAM :



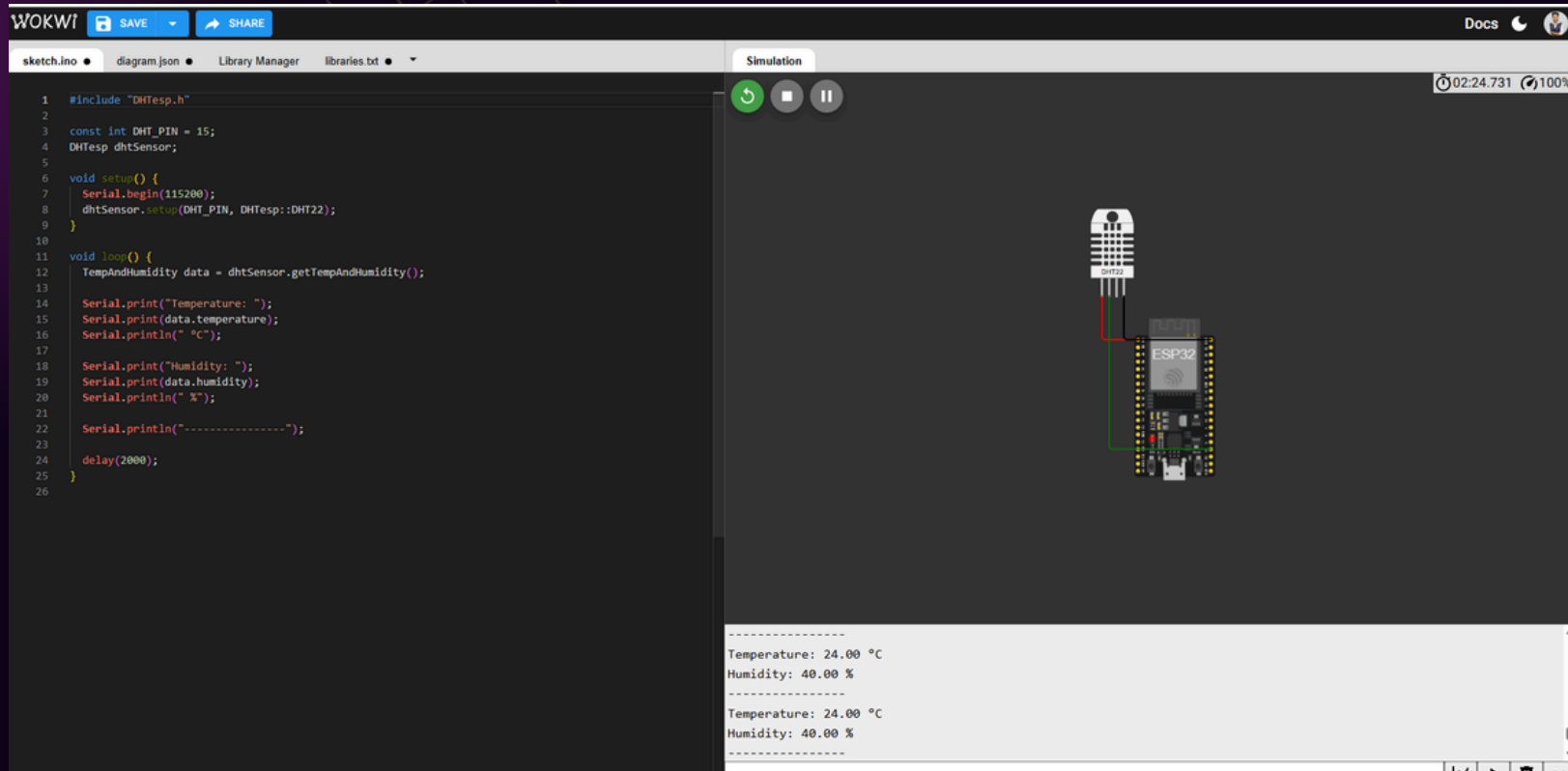
The DHT22 sensor is connected to the ESP32 microcontroller. The sensor measures temperature and humidity values and sends the data to the ESP32 for processing.

PROGRAM CODE :



The program reads temperature and humidity data from the DHT22 sensor and displays the values through the Serial Monitor.

OUTPUT :



- Temperature: 24.00°C
- Humidity: 40.00%

The simulation successfully displayed real-time temperature and humidity values. The data was continuously monitored and logged through the Serial Monitor.

CONCLUSION & SKILLS LEARNED :

This task demonstrated how sensor data can be collected, displayed, and monitored using an IoT simulation environment. The DHT22 sensor successfully measured temperature and humidity values and transmitted them to the ESP32.

Key Skills Learned :

- Sensor Data Collection
- ESP32 Basics
- Temperature & Humidity Monitoring
- Data Logging
- IoT Simulation
- Hardware-Software Integration